STEP-BY-STEP IMAGE PRE-PROCESSING IN FIJI – NFH-MASKED TDP43

Timing: ~15–20 min for parameter selection; ~10–30 s per file for automated processing

This workflow outlines the analysis pipeline for the identification and quantification of TDP-43 (TDP) localization within NFH structures. The process consists of two primary stages: image pre-processing in Fiji/ImageJ to generate masks and object identification in CellProfiler.

All scripts and pipelines can be found at the GitHub repository

Stage 1: Image Pre-processing in Fiji/ImageJ

1. Open Fiji and run the threshold selection macro ("Choose threshold.ijm") on the input image folder. This macro opens a random subset of images to allow the user to determine optimal thresholding parameters.

   a. Select Plugins > Macros > Run....

   b. Select the macro "Choose threshold.ijm" in the popup window and click Open.

   c. In the prompt window, input the channel number corresponding to the channel of interest (e.g., enter 1 if TDP is the first channel).

   d. Select the input folder containing the raw .ims images.

   e. Enter the number of random images to open for testing (the maximum available count will be displayed).

   f. Wait until the selected images are opened, channels are split, and the channel of interest is displayed.

2. Determine the optimal threshold method and range.

   a. Select Image > Adjust > Threshold... (or press Shift + Control + T).

   b. Adjust the threshold method and range to capture the maximum amount of NFH signal while minimizing noise.

   c. CRITICAL: Ensure that the chosen parameters distinguish signal from background noise; if intensities are identical or similar, segmentation may fail.

   d. Record the Threshold Method, Threshold Minimum, and Threshold Maximum values.

   e. Close all images (Press Shift + W).

f. Repeat steps 1a to 2e for the NFH channel

Note: This macro is intended for visual inspection and parameter optimization only; no output files are generated. Use this step to identify and record the threshold method, minimum, and maximum values required for the subsequent processing macro. This step may be run as many times as necessary/desired.

3. Run the processing macro ("Mask channels.ijm") to generate a masked molecule image. This macro applies the parameters determined in Step 2 to the entire dataset.

   a. Select Plugins > Macros > Run....

   b. Select the macro "Mask channels.ijm" and click Open.

   c. In the prompt window, input the name of the channel to be used as a mask (e.g., NFH) and the target molecule name (e.g., TDP).

   d. For each channel, enter the parameters recorded in Step 2d:

   i. Channel Number.

   ii. Threshold Method, Min, and Max.

   e. Enter the Mean Cutoff value (default is 0.95).

   f. Click OK to run the script.

   g. This will output one folder named "Channel_X_MASKNAME" (e.g., "Channel_1_NFH"), one folder named “Channel_X_MOLECULENAME (e.g., TDP) and a "pixel metadata" file containing pixel dimensions (µm²).

   Note: The **Mean Cutoff** parameter is a quality control step. It calculates the mean intensity (0 to 1) for each slice. Slices with a mean intensity higher than the set value (e.g., 0.95) are automatically deleted to remove artifacts where thresholding is calculated incorrectly. The default value is typically sufficient to ensure the removal of artifact slices.

4. Run the macro “Remerge Channels.py” in order to remerge the split images from the previous steps into one multichannel image

   a. Select Plugins > Macros > Run....

   b. Select the macro "Remerge channels.py" and click Open.

   c. In the popup window, select the folder containing the output from the previous step

   d. Mark the channels you want to include in the output image (e.g., NFH and

TDP)

e. Select the order in which the channels should be organized. (Suggestion: 1 – TDP; 2 – NFH)

Output: A new folder named "merged_[folder name]" will be created containing the final composite images.

Analysis using CellProfiler

Timing: ~5-10 min to set optimal parameters; ~10-20s to analyze each file

This section outlines the analysis pipeline for the identification and quantification of intact and degenerated NFH. The pipeline utilizes distinct modules for channel separation, object segmentation, and morphological measurement.

1. Initialize the CellProfiler project and load the pipeline.

   a. Open CellProfiler.

   b. Import the pipeline file “MoleculeInsideNFH.cppipe” by dragging and dropping it into the pipeline panel.

   c. Select the "Images" tab and drag and drop the pre-processed image folders (from the previous section) into the file list.

2. Configure output directory.
   a. Select "View output settings" to define the destination directory for the output images and for the generated .csv spreadsheets.

3. Optimize segmentation parameters in Test Mode. Select "Start Test Mode" to begin stepwise optimization.

Table 1: Cell Profiler modules

Module	Function	Action Required
Images	Select images to undergo analysis	Upload images from the previous pre-processing step
Metadata	Option to extract metadata	Not necessary to extract metadata
NamesAndTypes	Identify and name images	Certify that the assigned name is “nfh_ptdp43”
Groups	Option to group images	Not necessary to extract metadata
ColorToGray	Converts color images to grayscale	Make sure “Channel number” matches the chosen order from step 4e – Remerge channels
IdentifyPrimaryObjects	Identifying objects from a binary image for subsequent calculation	Adjust “Typical diameter of objects” until TDP is selected
ConvertObjectsToImage	Convert set of identified NFH objects to static image	No action required

SaveImages	Saving the previously created image to output path	Change file name under “Enter file prefix” if desired
IdentifyPrimaryObjects	Identifying objects from a binary image for subsequent calculation	Adjust “Typical diameter of objects” until NFH is selected
ConvertObjectsToImage	Converting set of identified degenerated NFH objects to static image	No action required
SaveImages	Saving the previously created image to output path	Change file name under “Enter file prefix” if desired
MaskObjects	Removing objects outside a speficic region defined by a mask	No action required
ConvertObjectsToImage	Converting set of masked TDP objects to static image	No action required
SaveImages	Saving the previously created image to output path	Change file name under “Enter file prefix” if desired
OverlayOutlines	Placing identified objects over a binary image for ease of visualization	Change colors as desired
SaveImages	Saving the previously created image to output path	Change file name under “Enter file prefix” if desired
MeasureImageAreaOccupied	Calculating area of identified objects	No action required
MeasureObjectSizeShape	Calculating number of identified objects	No action required
ExportToSpreadsheet	Exporting results to .csv files in the output path	No action required

4. Run the analysis.

   a. Once settings are optimized in Test Mode, select "Exit Test Mode".

   b. Select "Analyze Images" to execute the pipeline on the full dataset.

   c. Upon completion, the data will be exported as .csv files (e.g., "Image.csv") in the designated output directory.

5. **Note:** For statistical analysis, the "AreaOccupied" and "Count" columns in the output spreadsheet should be used. To convert pixel area to µm², multiply the pixel count by the pixel area value located in the "pixel metadata" file

generated during the pre-processing stage. Objects that are not segmented properly can be identified by reviewing the output overlay images and should be excluded from downstream analysis.